

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T15 · Direct & Inverse Proportion

With Answers

Questions with full worked solutions

■ 25 pages

FAMILY MAP

Direct Inverse Proportion

F01

Build an inverse-variation model

8 Q | 25 marks

F02

Build an algebraic direct-variation model

3 Q | 12 marks

18 y is inversely proportional to x^3

$$y = 4 \text{ when } x = 3$$

Work out the value of x when $y = 864$

$$x = \dots\dots\dots$$

(Total for Question 18 is 4 marks)

Worked Solution 11

Inverse proportionality

UNIT 2**4 marks****Find the constant**

$$y = \frac{k}{x^3}, \quad 4 = \frac{k}{3^3} \Rightarrow k = 108.$$

Use y=864

$$864 = \frac{108}{x^3} \Rightarrow x^3 = \frac{1}{8} \Rightarrow x = \frac{1}{2}.$$

Final answer

$$x = \frac{1}{2}$$

Question 22

4MA1-2H Q17

ORIGINAL QUESTION

3 marks

17 Q is directly proportional to the square root of d

$$Q = 4.5 \text{ when } d = 324$$

Find a formula for Q in terms of d

Worked Solution 22

Direct proportion

MODULAR UNIT 2

3 marks

Constant

$$Q = k\sqrt{d}, \quad 4.5 = 18k \Rightarrow k = 1/4.$$

Final answer

$$Q = \frac{\sqrt{d}}{4}$$

Question 26

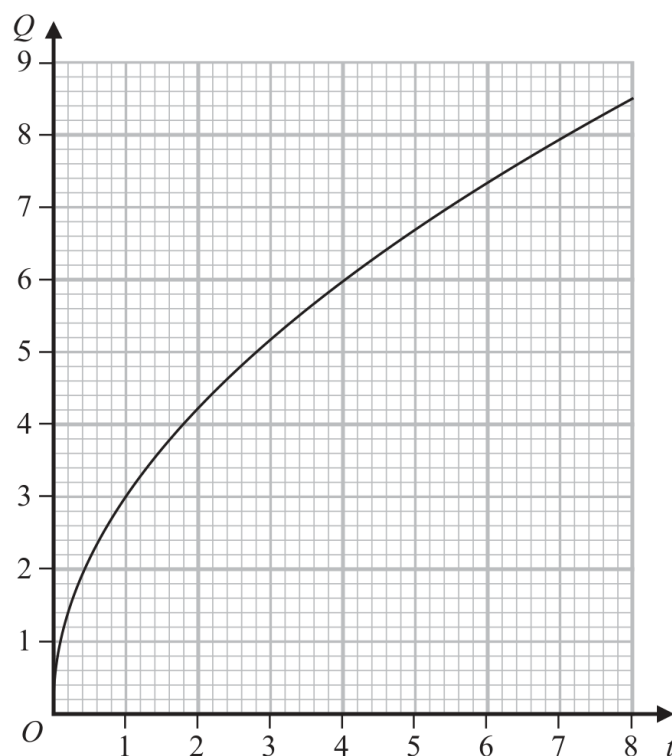
4MA1-2H Q16

ORIGINAL QUESTION

5 marks

16 Q is directly proportional to $\sqrt{t}$

The graph shows the relationship between Q and t for $0 < t < 8$



(a) Find a formula for Q in terms of t

.....
(3)

Q is increased by 20%

(b) Find the percentage increase in t

..... %
(2)

Worked Solution 26

Root proportionality and percentage change

MODULAR UNIT 2

5 marks

Read the graph

$$Q \propto \sqrt{t}, \quad Q = k\sqrt{t}; \text{ a point such as } (1, 3) \text{ gives } k = 3.$$

Write the model

$$Q = 3\sqrt{t}.$$

Apply the 20% change

$$Q' = 1.2Q \Rightarrow 3\sqrt{t'} = 1.2(3\sqrt{t}) \Rightarrow t' = 1.2^2 t = 1.44t.$$

Convert to a percentage

$$(1.44 - 1) \times 100\% = 44\%.$$

Final answer

$$Q = 3\sqrt{t}; \quad 44\%$$

Original question 11

4MA1-1H Q14 M01

MODULAR UNIT 2

3 marks

14 B is inversely proportional to the square of d

$$B = 0.25 \text{ when } d = 12$$

Find a formula for B in terms of d

.....

Worked solution 11

Model inverse-square variation

MODULAR UNIT 2

3 marks

M01 – Model inverse-square variation**Write the model**

$$B = k/d^2.$$

Use the given pair

$$0.25 = k/12^2 \Rightarrow k = 36.$$

State the formula

$$B = 36/d^2.$$

Final answer

$$B = \frac{36}{d^2}$$

Original question 29

4MA1-2H Q17 Q17

ORIGINAL QUESTION

4 marks

- 17 M varies directly as the cube of h
 $M = 4$ when $h = 0.5$

Find the value of h when $M = 500$

Worked solution

Family : Build an algebraic direct – variation model • Variant : Power direct variation $y = kx^n$

METHOD / WORKING

MODULAR UNIT 2 • JUNE 2022 DETAILED SOLUTIONS
Worked solution 29 MODULAR UNIT 2
Chapter: Direct & Inverse Proportion 4 marks
Q17 – Chapter: Direct & Inverse Proportion
Family : Build an algebraic direct – variation model • Variant : Power direct variation $y = kx^n$
Method
Direct variation gives $M = kh^3$.
Why: Translate the variation statement into an equation.
Method
$4 = k(0.5)^3$, so $k = 32$.
Why: Use the given pair to determine the constant.
Method
$500 = 32h^3$, so $h^3 = 15.625$.
Why: Substitute the new M value and isolate h cubed.
Method
$h = \text{cube root of } 15.625 = 2.5$.
Why: Take the real cube root.
Final answer
h: $h = 2.5$

FINAL ANSWER

Original question 10

4MA1-1H Q15

ORIGINAL QUESTION**3 marks**

15 A is inversely proportional to C^2

$A = 40$ when $C = 1.5$

Calculate the value of C when $A = 1000$

$C = \dots\dots\dots$

Worked solution

Family: Build an inverse-variation model • Variant: Recover the input under inverse variation

METHOD / WORKING

MODULAR UNIT 2 • JANUARY 2022 DETAILED SOLUTIONS
Worked solution 10 MODULAR UNIT 2
Chapter: Direct & Inverse Proportion 3 marks
Q15 – Chapter: Direct & Inverse Proportion
Family: Build an inverse-variation model • Variant: Recover the input under inverse variation
Method
$A = k/C^2$ and $40 = k/(1.5)^2$, so $k = 90$.
Why: Use inverse proportionality to find the constant.
Method
When $A = 1000$, $1000 = 90/C^2$, so $C^2 = 0.09$.
Why: Substitute the new value of A.
Method
$C = 0.3$ (taking the positive value for this quantity).
Why: Take the positive square root.
Final answer
C: 0.3

FINAL ANSWER

16 A is inversely proportional to the square of r

$$A = 5 \text{ when } r = 0.3$$

(a) Find a formula for A in terms of r

.....
(3)

(b) Find the value of A when $r = 7.5A$

$A =$
(3)

Worked solution

June 2021 | Question 16 | 3 marks

Family: Build an inverse-variation model | Variant: Inverse-square or inverse-power variation

METHOD / WORKING

1. Write inverse – square proportionality as $A = k/r^2$.

2. Use $A=5$ when $r=.3$: $5=k/(\text{.3})^2$, so $k=.45$.

3. Therefore $A = .45/r^2$.

FINAL ANSWER

$$A = 0.45/r^2$$

16 A is inversely proportional to the square of r

$$A = 5 \text{ when } r = 0.3$$

(a) Find a formula for A in terms of r

.....
(3)

(b) Find the value of A when $r = 7.5A$

$A =$
(3)

Worked solution

June 2021 | Question 16 | 3 marks

Family: Build an inverse-variation model | Variant: Inverse-square or inverse-power variation

METHOD / WORKING

1. *Substituter* $= 7.5A$ into $A = .45/r^2$.

2. *Rearrange* : $A^3 = .45/(7.5^2) = .008$.

3. Take the positive cube root: $A = .2$.

FINAL ANSWER

$$A = 0.2$$

14 F is inversely proportional to the square of v .

Given that $F = 6.5$ when $v = 4$

find a formula for F in terms of v .

.....
(Total for Question 14 is 3 marks)

Answer reference | May-Nov 2020 | Question 14

Family: Build an inverse-variation model • Variant: Inverse-square or inverse-power variation

14	$F = \frac{k}{v^2}$ or $Fv^2 = k$ oe		3	M1 (NB. Not for $F = \frac{1}{v^2}$) Constant of proportionality must be a symbol such as k	M2 for $6.5 = \frac{k}{4^2}$ oe
	$6.5 = \frac{k}{4^2}$ or $k = 6.5 \times 4^2$ or $k = 104$			M1 For substitution of F and v into a correct formula	
		$F = \frac{104}{v^2}$		A1 Award 3 marks if $F = \frac{k}{v^2}$ is on the answer line and the value of $k = 104$ is found	
Total 3 marks					

Worked solution

May-Nov 2020 | Question 14 | 3 marks

Family: Build an inverse-variation model | Variant: Inverse-square or inverse-power variation

METHOD / WORKING

Worked solution

May-Nov 2020 | Question 14 | 3 marks

Family: Build an inverse-variation model | Variant: Inverse-square or inverse-power variation

Method

F is inversely proportional to v^2 , so

$$F = (k)/(v^2)$$

Use $F=6.5$ when $v=4$:

$$6.5 = (k)/(4^2)$$

$$k = 6.5 \times 16 = 104$$

Therefore

$$F = (104)/(v^2)$$

Answer: $F = (104)/(v^2)$.

FINAL ANSWER

17 y is inversely proportional to $\sqrt{x}$

$y = c^4$ when $x = c^2$ where c is a positive constant.

Find a formula for y in terms of x and c

Give your answer in its simplest form.

(Total for Question 17 is 3 marks)

Answer reference | January 2023 | Question 17

Family: Build an inverse-variation model • Variant: Inverse-square or inverse-power variation

17	$y = \frac{k}{\sqrt{x}}$ or $ky = \frac{1}{\sqrt{x}}$ or $\sqrt{x} = \frac{k}{y}$ oe		3	M1 (NB. Not for $y = \frac{1}{\sqrt{x}}$) Constant of proportionality must be a symbol such as k (Allow c for k for this mark only)	M2 for $c^4 = \frac{k}{\sqrt{c^2}}$ oe
	$c^4 = \frac{k}{\sqrt{c^2}}$ oe or $k = c^4 \times \sqrt{c^2}$ oe			M1 for substitution of x and y into a correct formula	
	<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	$y = \frac{c^5}{\sqrt{x}}$		A1 oe e.g. $y = c^5 \times \frac{1}{\sqrt{x}}$ Award 3 marks if answer is $y = \frac{k}{\sqrt{x}}$ on the answer line and $k = c^5$ clearly given in the body of working of the script	
				Total 3 marks	

Worked solution

January 2023 | Question 17 | 3 marks

Family: Build an inverse-variation model • Variant: Inverse-square or inverse-power variation

Official final mark-scheme pages are embedded immediately after this question.

17 T is inversely proportional to $\sqrt{m}$

$$T = 15 \text{ when } m = 36$$

Find a formula for T in terms of m

(Total for Question 17 is 3 marks)

Worked solution

November 2025 | Question 17 | 3 marks

Family: Build an inverse-variation model • Variant: Inverse-square or inverse-power variation

Method

T is inversely proportional to m, so

$$T = (k)/(m)$$

Use $T=15$ when $m=36$:

$$15 = (k)/(36)$$

$$k = 540$$

Therefore

$$T = (540)/(m)$$

Answer: $T = (540)/(m)$.